

Week 2: Data Acquisition and Preliminary Analysis

Dataset: Titanic Dataset

1. Introduction

The objective of this task is to understand the complete process of acquiring a dataset, studying its structure, performing data quality checks, and conducting preliminary exploratory analysis.

Data acquisition and preliminary analysis are important steps in any data science project because they help researchers understand the available information before applying machine learning algorithms.

In this task, the Titanic dataset was selected and analyzed using Python and Google Colab. The analysis focused on understanding passenger details, survival patterns, missing data, variable types, and relationships between different features.

2. Dataset Selection

Dataset Name: Titanic Dataset

Source:

The Titanic dataset is a publicly available dataset commonly used for data analysis and machine learning practice. It is available through Kaggle and other open-source platforms.

Number of Records:

The dataset contains 891 passenger records and 12 attributes.

Important Variables:

PassengerId:

A unique identification number assigned to every passenger.

Survived:

Target variable indicating whether a passenger survived.

0 represents not survived and 1 represents survived.

Pclass:

Represents passenger class:

1 - First Class

2 - Second Class

3 - Third Class

Sex:

Represents passenger gender.

Age:

Shows the age of passengers.

SibSp:

Number of siblings or spouses travelling with the passenger.

Parch:

Number of parents or children travelling with the passenger.

Fare:

Ticket price paid by passengers.

Cabin:

Cabin number information.

Embarked:

Port from where passengers boarded the ship.

Reason for Selecting Dataset:

The Titanic dataset was selected because it contains real-world data, missing values, numerical and categorical variables, and meaningful patterns. It is suitable for practicing data preprocessing and exploratory data analysis.

3. Data Acquisition Strategy

Dataset Obtaining Process:

The Titanic dataset was obtained from a public data repository in CSV format. CSV format is widely used because it is easy to store, share, and process using Python.

Downloading Dataset:

1. The dataset file was downloaded from the online repository.
2. The CSV file was uploaded into Google Colab.
3. Python Pandas library was used to read the dataset.

Loading Dataset in Google Colab:

Example Code:

```
from google.colab import files
uploaded = files.upload()
```

```
import pandas as pd
df = pd.read_csv('titanic.csv')
```

Data Storage:

After loading, the dataset was stored in a Pandas DataFrame called df. DataFrame format allows easy data manipulation, filtering, cleaning, and visualization.

4. Data Understanding

Data understanding is the process of examining the dataset structure and identifying important characteristics.

df.head():

Displays the first five rows of the dataset and helps understand column values.

df.shape:

Shows the number of rows and columns available in the dataset.

Output:

(891, 12)

df.info():

Provides information about column names, data types, and missing values.

df.describe():

Generates statistical information such as mean, standard deviation, minimum, maximum, and quartile values.

Data Types:

Numerical Data:

- PassengerId
- Survived
- Pclass
- Age
- SibSp
- Parch
- Fare

Categorical Data:

- Name
- Sex
- Ticket

- Cabin
- Embarked

5. Data Cleaning

Data cleaning is required to improve dataset quality and remove problems before analysis.

Missing Age Values:

The Age column contains missing values. Since age is an important feature, missing values are replaced using the median value of available ages.

Missing Embarked Values:

The Embarked column contains a few missing values. These are replaced using the most frequently occurring category.

Cabin Column:

The Cabin column contains many missing values. Due to a large number of missing entries, it requires removal or transformation into useful features.

Duplicate Records:

Duplicate records are checked to avoid repeated information affecting analysis.

Categorical Consistency:

Categorical columns such as Sex and Embarked are checked to ensure values are consistent.

Outlier Investigation:

Numerical columns such as Age and Fare are analyzed to identify unusual values. High fare values are checked because they may influence statistical analysis.

6. Python Code

Important Python Libraries:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Loading Dataset:

```
df = pd.read_csv('titanic.csv')
```

Data Checking:

```
df.head()
df.shape
```

```
df.info()  
df.describe()
```

Missing Value Analysis:

```
df.isnull().sum()
```

Data Cleaning:

```
df['Age'].fillna(df['Age'].median(), inplace=True)  
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

7. Exploratory Data Analysis

Exploratory Data Analysis (EDA) helps discover hidden patterns and relationships.

Survival Distribution:

The survival count is analyzed to understand the number of passengers who survived and did not survive.

Gender vs Survival:

Comparison between gender and survival shows differences in survival probability.

Passenger Class vs Survival:

Passenger class is analyzed to identify whether social class affected survival chances.

Age Histogram:

Shows the distribution of passenger ages.

Fare Histogram:

Displays ticket fare distribution and identifies high-value fare observations.

Age vs Fare Scatter Plot:

Used to understand whether age and fare have any relationship.

Correlation Analysis:

A correlation matrix is created to measure relationships between numerical variables.

8. Initial Findings

The analysis produced the following observations:

1. Survival rates were different between male and female passengers.
2. Female passengers had higher survival probability compared to male passengers.

3. Passenger class had an important relationship with survival.
4. First-class passengers had better survival chances.
5. The dataset contained missing values requiring preprocessing.
6. Age values showed a wide distribution.
7. Fare values contained some extreme high-value observations.
8. Numerical features showed different correlation levels.

9. Conclusion

In this task, the Titanic dataset was successfully acquired, loaded, and analyzed using Python in Google Colab.

The dataset structure, data types, missing values, and statistical properties were studied. Data cleaning techniques were applied to improve data quality.

Exploratory Data Analysis provided important insights about passenger survival patterns. This preliminary analysis will support future activities such as feature engineering and machine learning model development.

10. References

1. Kaggle Titanic Dataset
2. Python Official Documentation
3. Pandas Documentation
4. Google Colab Documentation
5. Matplotlib Documentation
6. Seaborn Documentation

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